THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 13: **Our Environment** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Substances which can be broken down by microorganisms are 1

(a) inorganic only **(b)** non-biodegradable
(c) biodegradable **(d)** synthetic

2. An artificial ecosystem is 1

(a) a pond **(b)** a forest **(c)** a lake **(d)** a crop field

3. A network of many interconnected food chains is called 1

(a) a habitat **(b)** a pyramid **(c)** a niche **(d)** a food web

4. The increase in concentration of harmful chemicals at successive trophic levels is called 1

(a) biomagnification **(b)** eutrophication **(c)** photosynthesis **(d)** decomposition

5. A natural ecosystem is 1

(a) a crop field **(b)** a forest **(c)** a garden **(d)** an aquarium

6. If 10,000 J of energy is available at the producer level, the energy available to the secondary consumer is 1

(a) 10,000 J **(b)** 1000 J **(c)** 100 J **(d)** 10 J

7. Which trophic level has the highest concentration of harmful chemicals like pesticides? 1

(a) Top carnivore **(b)** Decomposer **(c)** Herbivore **(d)** Producer

8. The ultimate source of energy for all ecosystems is 1

(a) the Sun **(b)** wind **(c)** water **(d)** the Moon

9. Number of trophic levels in a food chain is usually limited to 3 or 4 because 1

(a) predators are few **(b)** plants are few
(c) energy is lost at each step **(d)** food is not available

10. Use of dustbins, recycling and composting are methods of 1

(a) food preservation **(b)** air purification
(c) waste management **(d)** water purification

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Differentiate between biodegradable and non-biodegradable waste. 2

12. Define a food chain. Give an example. 2

13. What are CFCs? Name one use. 2

14. Why should we avoid using plastic bags? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the energy available to the third trophic level if the producers have 20,000 J of energy. 3

16. How does the ozone layer get damaged? State its effects. 3

17. What are the harmful effects of plastics on the environment? 3

OR

Why are producers called the base of the food chain? 3

18. How can we reduce the damage to the ozone layer? 3

OR

Suggest three ways of reducing the waste produced at home. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the flow of energy in an ecosystem and why it is called unidirectional. 5

OR

How is ozone formed in the stratosphere and how is it destroyed by CFCs? Why is its depletion a concern? 5

20. Describe the effects of ozone depletion. Name the chemicals causing it and the steps taken to reduce it. 5

OR

Explain how human activities have affected the environment, with reference to plastics, CFCs, and pesticides. Suggest how each can be controlled. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND